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/*
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Project:
Dual-Wavelength Pulse Oximeter with OLED Display and Status LEDs
*/

// =====
// Library Includes
// =====
#include "MAX30100_PulseOximeter.h" // MAX30100 sensor library
#include <SPI.h>                     // SPI library (required by some
displays)
#include <Wire.h>                    // I2C communication library
#include <Adafruit_GFX.h>            // Graphics library
#include <Adafruit_SSD1306.h>        // OLED display library

// =====
// Constant Definitions
// =====
#define REPORTING_PERIOD_MS 1000    // Screen update interval (1 second)
#define SCREEN_WIDTH 96             // OLED width in pixels
#define SCREEN_HEIGHT 16            // OLED height in pixels
#define OLED_RESET -1               // OLED reset pin (not used)
#define SCREEN_ADDRESS 0x3C         // I2C address of OLED

// =====
// Object Declarations
// =====
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);
PulseOximeter pox;

// =====
// Global Variables
// =====
int menu = 0;                       // Display menu state (0-2)
int heartRate = 0;                  // Measured heart rate (BPM)

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int SpO2 = 0;                                // Measured blood oxygen saturation
(%)
uint32_t tsLastReport = 0;                   // Timestamp of last display update

// =====
// I/O Pin Definitions
// =====
int switchA = A0;                            // Menu decrement button
int switchB = A1;                            // Menu increment button
int ledGreen = A3;                           // Green status LED
int ledRed = A2;                             // Red status LED
int ledBdrA = D3;                            // Border LED A
int ledBdrB = A7;                            // Border LED B

// =====
// Debounce Variables
// =====
int buttonStateA;
int buttonStateB;
int lastButtonStateA = HIGH;
int lastButtonStateB = HIGH;
unsigned long lastDebounceTimeA = 0;
unsigned long lastDebounceTimeB = 0;
unsigned long debounceDelay = 100; // Debounce delay in ms

// =====
// Setup Function
// =====
void setup()
{
    Serial.begin(9600);
    Serial.println("Initializing pulse oximeter...");

    // Initialise MAX30100 sensor
    if (!pox.begin()) {
        Serial.println("MAX30100 init failed");
        while (1); // Halt if sensor fails
    }

    // Initialise OLED display

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if (!display.begin(SSD1306_SWITCHCAPVCC, SCREEN_ADDRESS)) {
    while (1);                // Halt if display fails
}

// Configure display settings
display.clearDisplay();
display.setTextSize(1);
display.setTextColor(SSD1306_WHITE);
display.setCursor(0, 0);
display.cp437(true);
display.display();

// Configure I/O pin modes
pinMode(switchA, INPUT);
pinMode(switchB, INPUT);
pinMode(ledGreen, OUTPUT);
pinMode(ledRed, OUTPUT);
pinMode(ledBdrA, OUTPUT);
pinMode(ledBdrB, OUTPUT);
}

// =====
// Main Loop
// =====
void loop()
{
    pox.update();              // Update sensor readings
    updateLEDS();              // Update LEDs based on heart rate

    // Update display at fixed interval
    if (millis() - tsLastReport > REPORTING_PERIOD_MS) {
        heartRate = pox.getHeartRate();
        SpO2 = pox.getSpO2();
        writeToScreen();
        tsLastReport = millis();
    }

    // Read button states
    int readingA = digitalRead(switchA);
    int readingB = digitalRead(switchB);

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// Reset debounce timers if state changes
if (readingA != lastButtonStateA) lastDebounceTimeA = millis();
if (readingB != lastButtonStateB) lastDebounceTimeB = millis();

// Handle button A (menu down)
if ((millis() - lastDebounceTimeA) > debounceDelay) {
    if (readingA != buttonStateA) {
        buttonStateA = readingA;
        if (buttonStateA == HIGH) menu--;
    }
}

// Handle button B (menu up)
if ((millis() - lastDebounceTimeB) > debounceDelay) {
    if (readingB != buttonStateB) {
        buttonStateB = readingB;
        if (buttonStateB == HIGH) menu++;
    }
}

lastButtonStateA = readingA;
lastButtonStateB = readingB;

// Constrain menu range
if (menu > 2) menu = 0;
if (menu < 0) menu = 2;
}

// =====
// Display Output Function
// =====
void writeToScreen()
{
    display.clearDisplay();

    switch (menu)
    {
        case 0: // Heart rate and SpO2
            display.setTextSize(1);

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        display.setCursor(0, 0);
        display.print("HR: ");
        display.print(heartRate);
        display.println(" bpm");
        display.setCursor(0, 8);
        display.print("SpO2: ");
        display.print(SpO2);
        display.println(" %");
        break;

    case 1:                                // Heart rate only
        display.setTextSize(2);
        display.setCursor(0, 0);
        display.print("HR:");
        display.print(heartRate);
        display.println("bpm");
        break;

    case 2:                                // SpO2 only
        display.setTextSize(2);
        display.setCursor(0, 0);
        display.print("SpO2:");
        display.print(SpO2);
        display.println("%");
        break;
}

display.display();
}

// =====
// LED Status Function
// =====
void updateLEDS()
{
    // Outside normal heart rate range
    if (heartRate > 110 || heartRate < 60)
    {
        digitalWrite(ledRed, HIGH);
        digitalWrite(ledGreen, LOW);
    }
}

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    digitalWrite(ledBdrA, HIGH);  
    digitalWrite(ledBdrB, LOW);  
}  
else  
{  
    digitalWrite(ledRed, LOW);  
    digitalWrite(ledGreen, HIGH);  
    digitalWrite(ledBdrA, LOW);  
    digitalWrite(ledBdrB, HIGH);  
}  
}
```